DATA ANALYTICS REPORT

EXPLORATORY DATA ANALYSIS OF YOUTUBE TRENDING VIDEOS DATASET

Content Trends, Viewership Dynamics & User Engagement

TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

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PROBLEM STATEMENT & OBJECTIVE

Problem Statement:

The YouTube platform hosts millions of videos across diverse categories. With massive variations in views, likes, and engagement, it becomes challenging to understand content trends, audience preferences, and what drives video success.

Objective:

To explore the YouTube trending videos dataset and uncover patterns, insights, and trends that help creators and businesses make data driven decisions.

EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01 Data Collection

- Gathered the Google Play Store dataset from Kaggle

02 Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03 Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04 Exploratory Analysis

- Univariate Analysis: e.g., Ratings, Installs, Prices
- Bivariate Analysis: Rating vs Reviews
- Multivariate Analysis: Reviews vs Rating vs Installs

05 Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

KEY QUESTIONS EXPLORED

- How does upload timing (day, hour) influence reach and engagement and when are the most effective publishing windows?
- Which content categories drive the highest reach & where does competition versus niche opportunity exist?
- Is consistent publishing more reliable than relying on rare viral hits for long-term performance?
- How are views, likes, comments & dislikes related & which engagement signals matter most for trending?
- Do regional differences affect engagement patterns and where is country-specific optimization most effective?
- How do platform policies and external factors influence video longevity beyond engagement metrics?

DATA OVERVIEW

TThe dataset provides key information about YouTube trending videos, including views, likes, comments, category, publish date, and trending date. Additional details such as channel information and country capture content performance, audience engagement and regional trends.

Data Source: Kaggle

Dataset Size

161470

Records

18

Features

App Diversity

56905

Videos covered

4

Countries

18

Video Categories

DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Index	Numerical (Discrete)	Dataset row number
Video_id	String	Unique YouTube video ID
Trending_date	Date	Date the video trended on YouTube
Title	String	Video title
Channel_title	String	Publishing channel name
Category_id	Numerical (Discrete)	YouTube category ID
Publish_date	Date	Video publication date
Time_frame	Categorical	Price of the app (0 indicates a free app)
Published_day_of_week	Date	Day of the week published
Publish_country	Categorical	Country of publication
Tags	String	The tags or keywords associated with the video.
Views	Categorical	The number of views received by a particular video
Likes	Numerical (Discrete)	The number of likes received by a particular video
Dislikes	Numerical (Discrete)	The number of dislikes received by a particular video
Comment_count	Numerical (Discrete)	The number of comments received by a particular video
Comments_disabled	Boolean	Whether comments are disabled
Ratings_disabled	Boolean	Whether ratings are disabled
Video_error_or_removed	Boolean	Whether the video is unavailable, removed, or errored

DATA QUALITY CHALLENGES & ANOMALIES

Few inconsistencies were found in the dataset, which could have affected the analysis if left unaddressed.

DATA ANOMALIES

- **Columns requiring cleaning or type conversion**: trending_date, publish_date
- **Columns with notable anomalies or invalid entries**:
    - tags (very long strings, |-separated values)
    - publish_country (may require standardization with short forms)

DATA CLEANING & TREATMENT

Inconsistent and missing values were addressed, and key features were cleaned and standardized for analysis.

DATA CLEANING SUMMARY

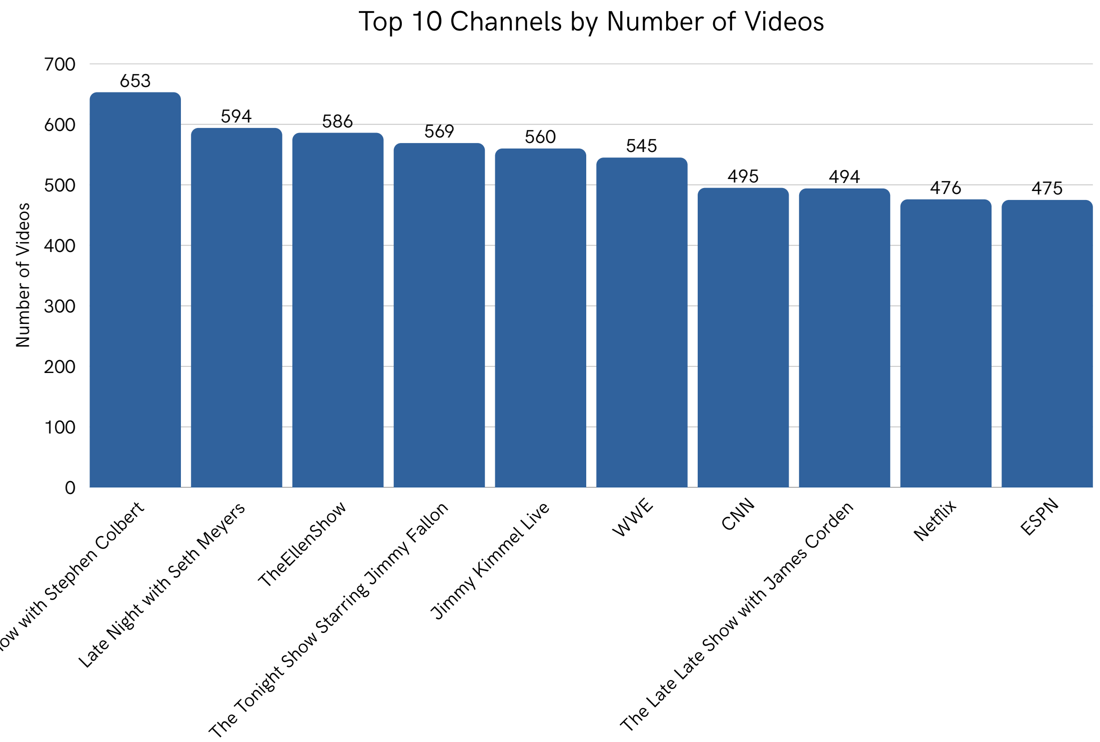
- Features such as **trending_date, publish_date** were standardized with respect to data type conversion.
- Short forms the **publish_country** feature were standardized.
- The feature **tags** was transformed into a clean, standardized strings by splitting, normalizing, deduplicating and removing irrelevant tags.
- New features **days_to_trend** & **category_name** were created from trending_date, publish_date & category_id respectively.
- **index and video_id** was removed as it is a technical detail with limited business relevance.

INSIGHTS

Number of Videos Published by Top Channels

5,447

Videos from Top 10 Channels



Key observations

- A small group of channels (late-night shows, major media brands) dominate the number of videos, while most creators appear far less frequently.

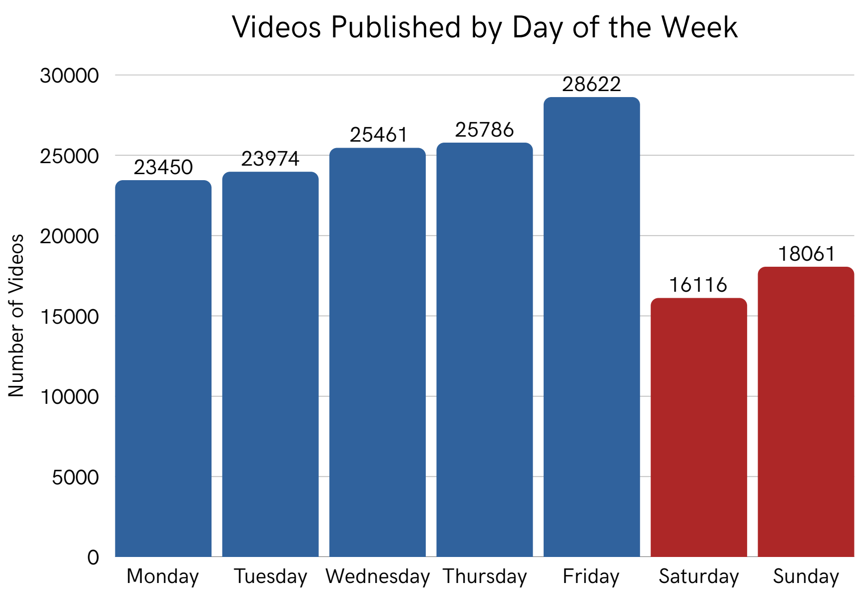
Business Insights

- Visibility is heavily concentrated among established brands.
- New or smaller channels need niche content or viral strategies to compete effectively.

Video Publishing Peaks on Weekdays and Drops on Weekends

23,067

Average Weekly Upload Volume



Key observations

- Video publishing peaks on weekdays, especially Friday and drops significantly over the weekend.

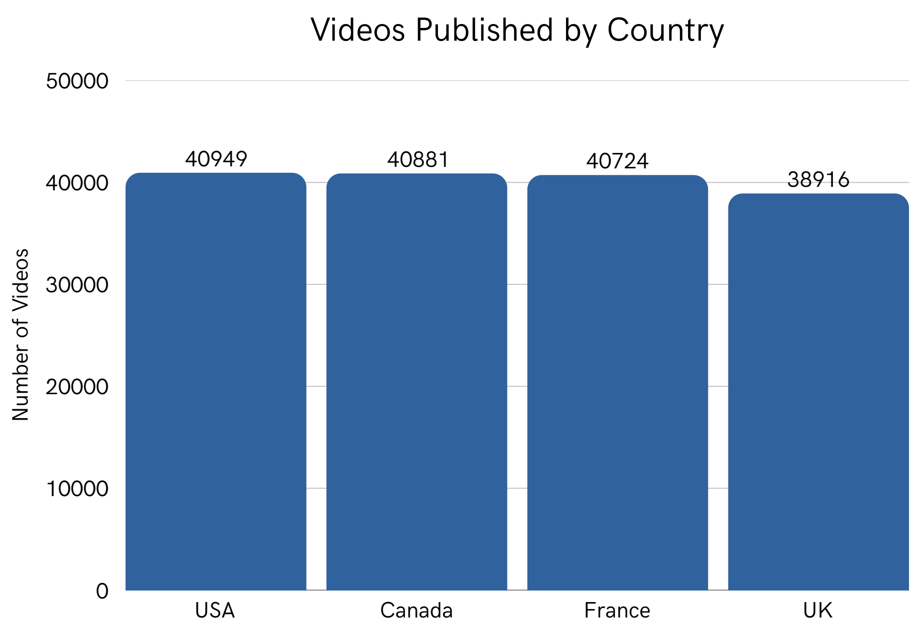
Business Insights

- Uploading during weekdays, particularly **Thursday–Friday** can improve visibility and engagement, while weekend uploads face lower competition but reduced overall activity.

Content Creation Is Evenly Distributed Across Major Countries

1,61,470

Total Videos from 4 Major Countries



Key observations

- Video publishing volume is fairly balanced across countries, with the USA, Canada and France contributing slightly more content than the UK.

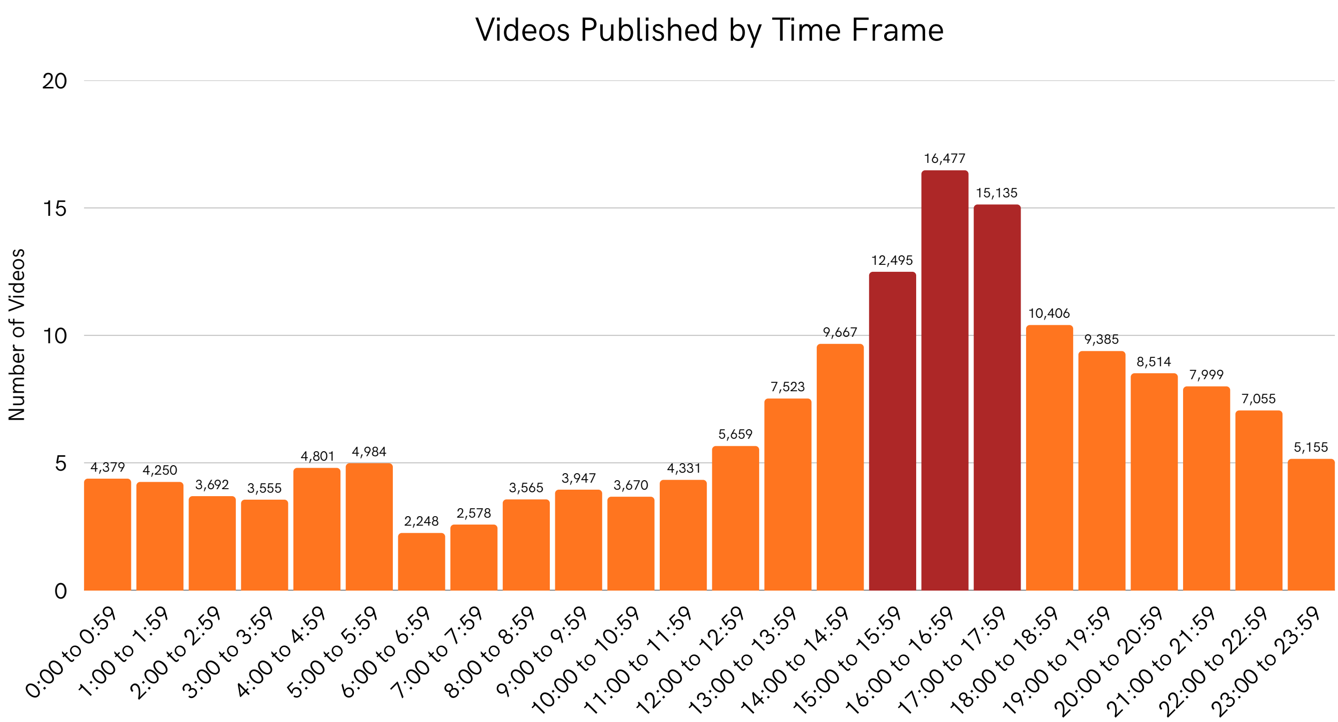
Business Insights

- The near-uniform distribution suggests global participation in content creation, so region-specific optimization (timing, language, trends) may be more effective than focusing on a single dominant market.

Most Videos Are Uploaded Between 3 PM and 6 PM

5069

Typical Upload Volume per Hour (Median)



Key observations

- Video uploads are heavily concentrated between 3 PM and 6 PM, with a clear peak around 4–5 PM.
- Early morning hours show the lowest publishing activity, indicating reduced creator presence.

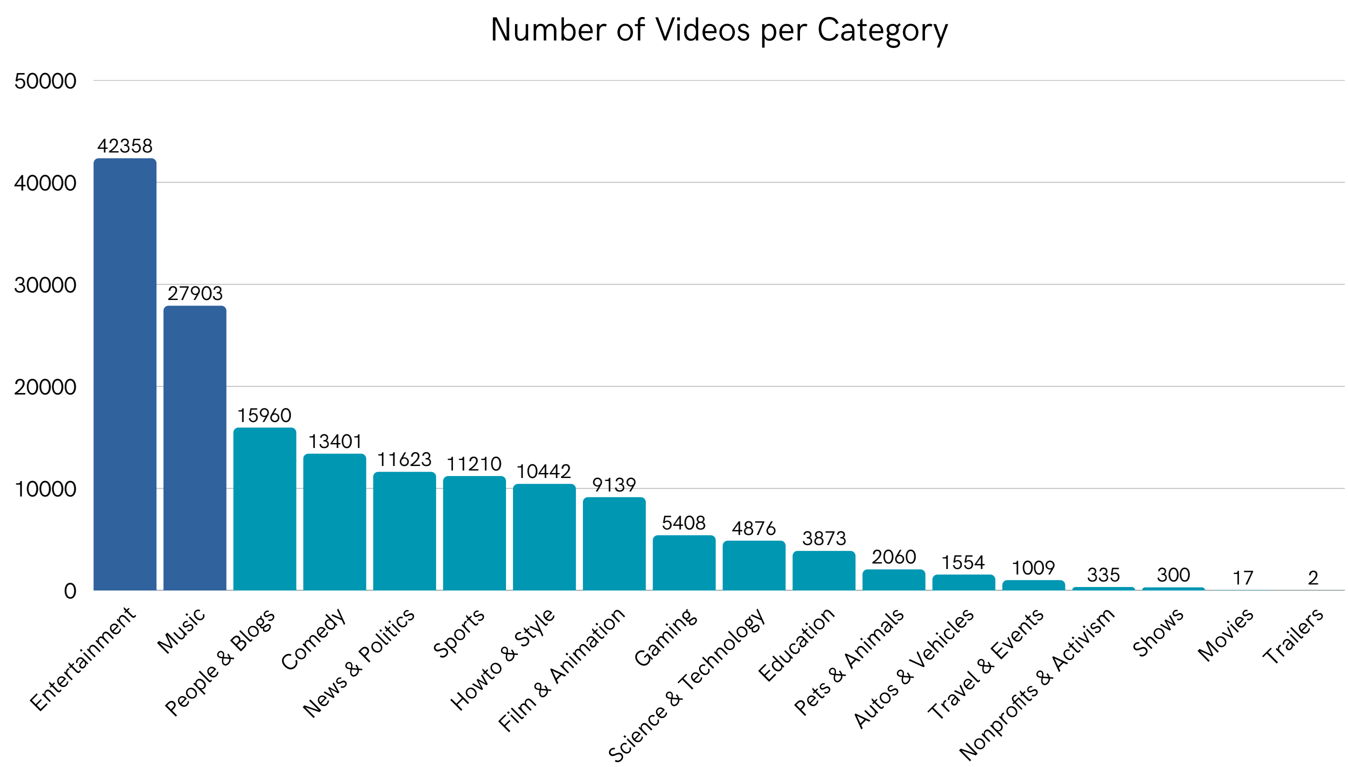
Business Insights

- Creators usually publish during peak audience hours, creating heavy competition, while late-night and early-morning windows may offer better visibility due to lower content saturation.

A Few Content Categories Dominate YouTube Upload Volume

43.51%

Of the Videos are in Music & Entertainment category



Key observations

- **Entertainment and Music** alone account for a disproportionately large share of total video uploads.
- Most other categories form a long tail with significantly lower content volume.

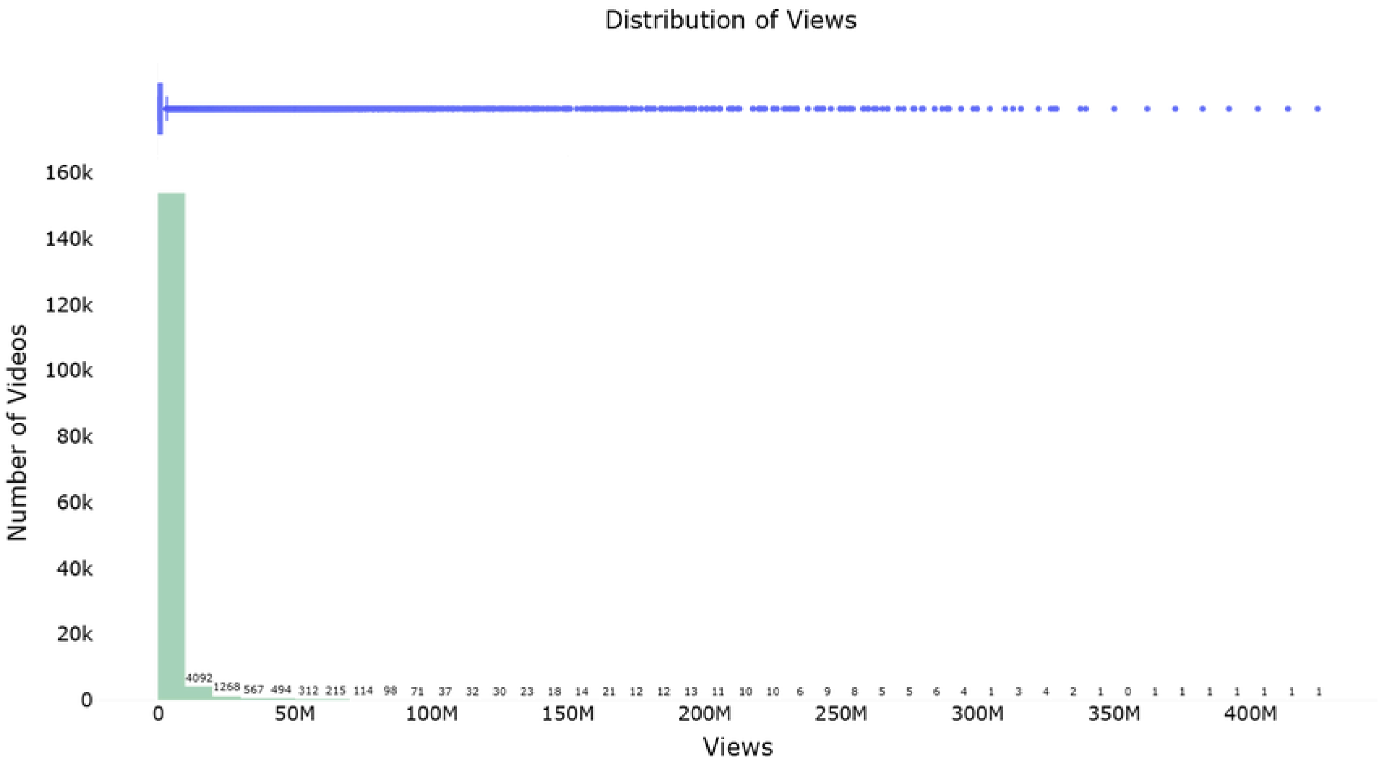
Business Insights

- High-volume categories face severe competition, making discoverability harder for new creators.
- Niche and under-served categories (eg. Education, Science & Technology) present opportunities for differentiation and targeted audience growth.

Most Videos Receive Modest Views, While Few Become Massive Hits

3,84,739

Typical Number of Views (Median)



Key observations

- Video views are extremely right-skewed, with the vast majority of videos receiving relatively low views and a small fraction achieving very high view counts.

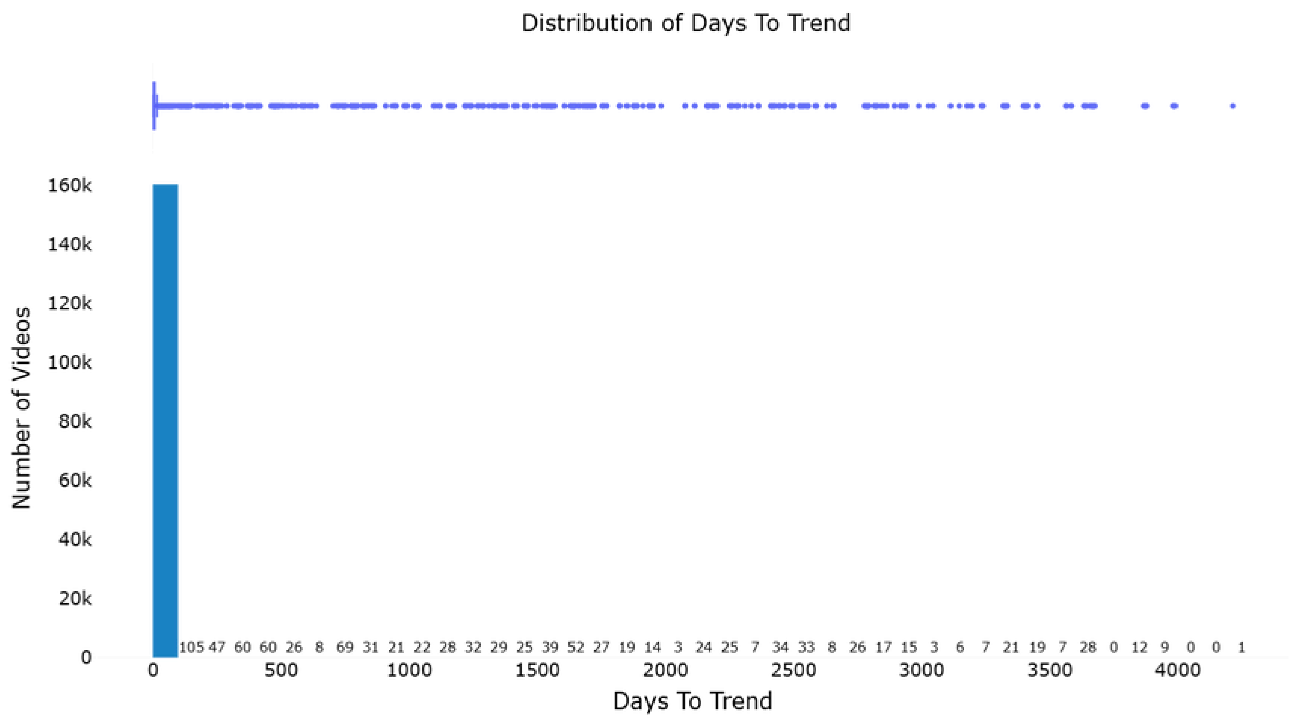
Business Insights

- Success is driven by rare viral hits, so relying on a few breakout videos is risky
- Consistent publishing and optimization are key to improving average performance over time.

Most Videos Trend Quickly,Late Breakouts Are Rare

3 Days

Median Time Taken to Trend



Key observations

- Most videos trend very quickly after publication, while a small number take a long time to appear on the trending list, creating a strong right-skewed distribution.

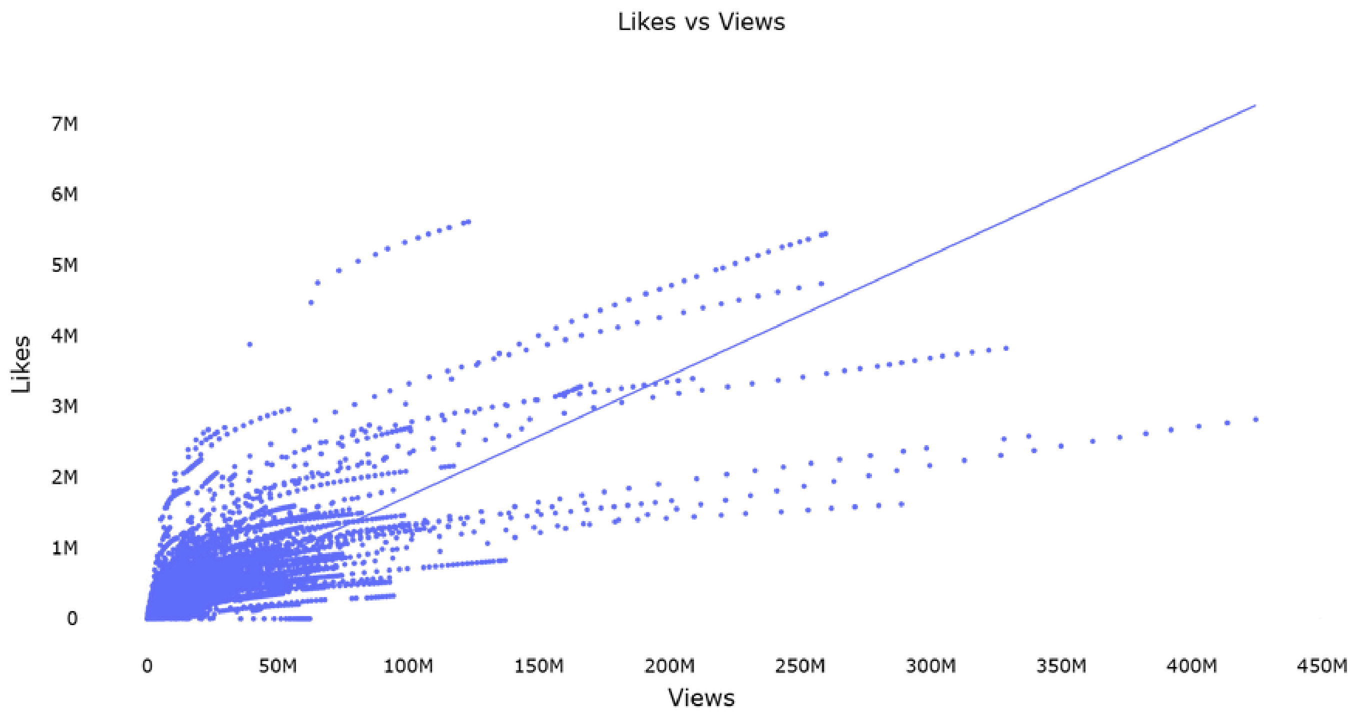
Business Insights

- Early performance is critical for trending success, so optimizing the first 24–48 hours (titles, thumbnails, timing, promotion) is far more impactful than relying on long-term discovery.

Higher Reach Drives Engagement, but Efficiency Varies Widely

9840

Typical Number of Likes (Median)



Key observations

- Likes increase with views, showing a clear positive relationship, though engagement efficiency varies significantly across videos at similar view levels.

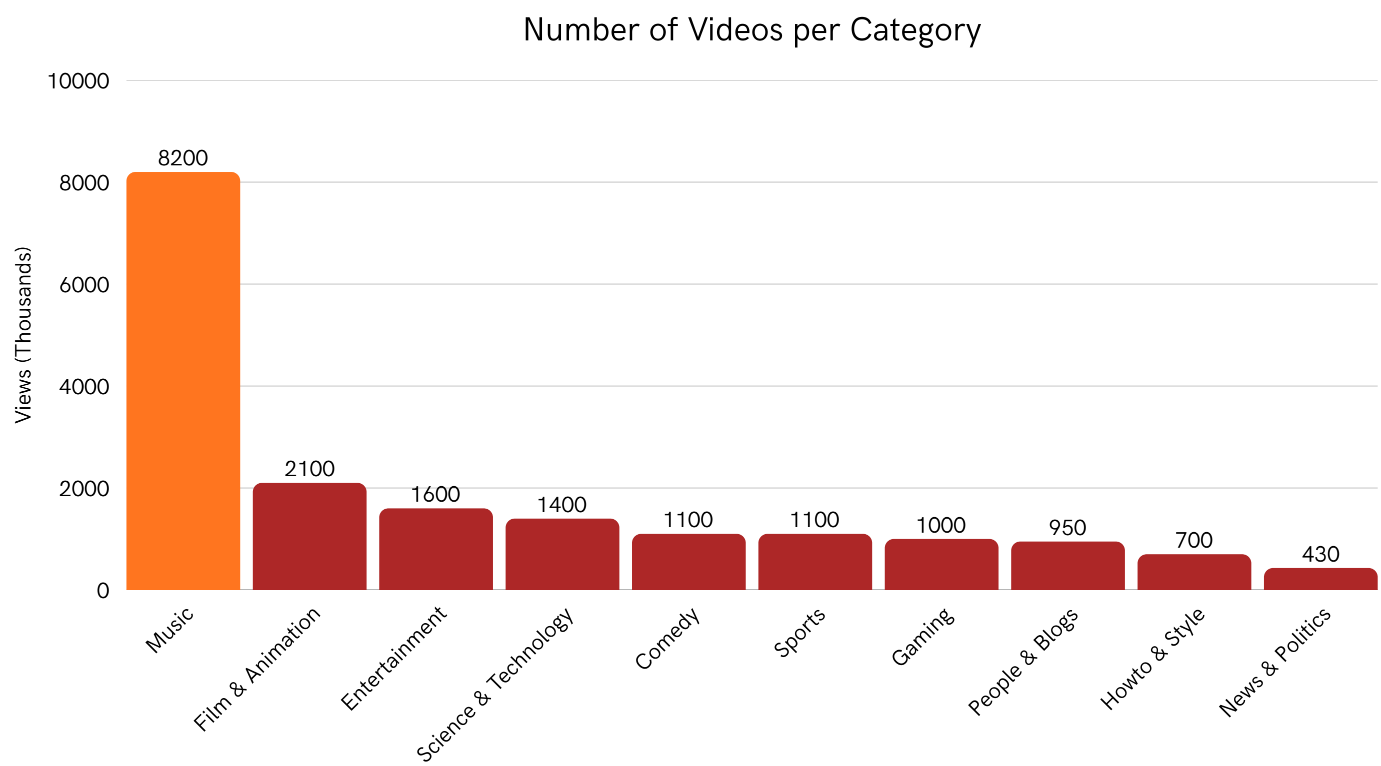
Business Insights

- While higher reach generally leads to more likes, optimizing content quality and audience relevance is key to improving like-to-view conversion rather than relying on views alone.

Music Dominates View Potential Across Categories

280K

Typical Number of Views per Category (Median)



Key observations

- Music videos have a significantly higher average view count than all other top categories, while News & Politics and How-to & Style attract comparatively lower average views.

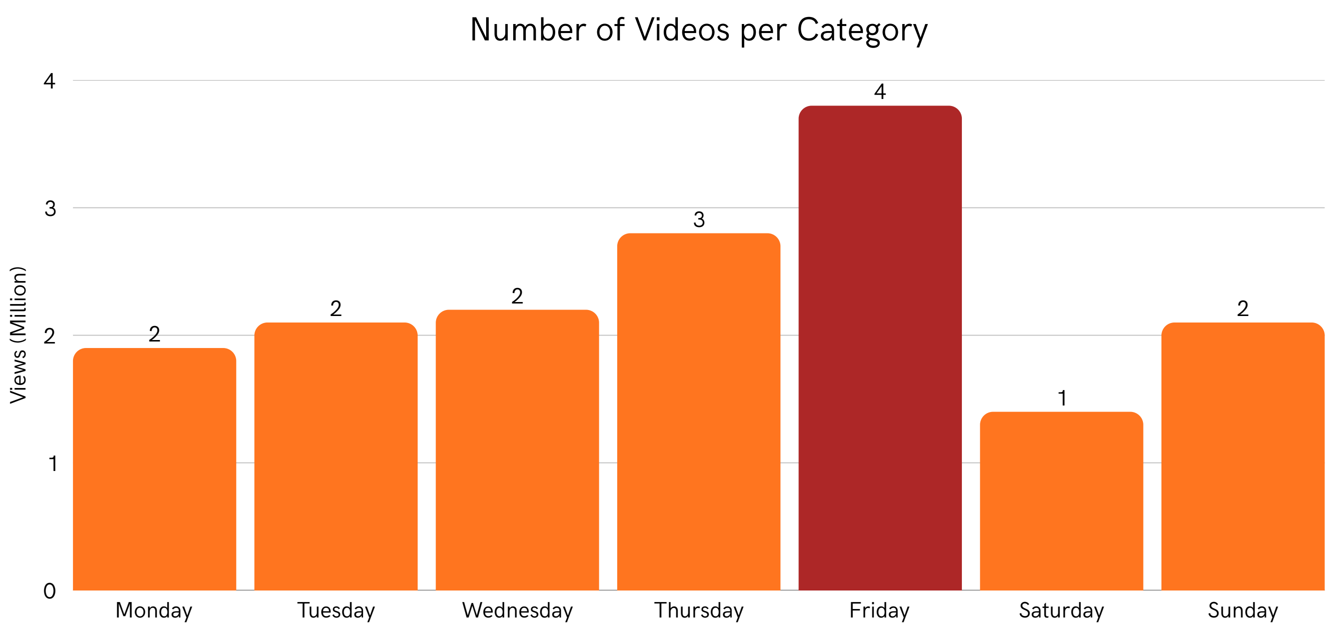
Business Insights

- Reach varies significantly by category, so content performance and audience size are not uniform across categories.
- Content and monetization strategies should be category-specific, high-reach categories like Music focus on mass exposure, while lower-view categories depend more on niche engagement and retention.

Friday Drives Maximum Views

2.3 M

Average Views per Week



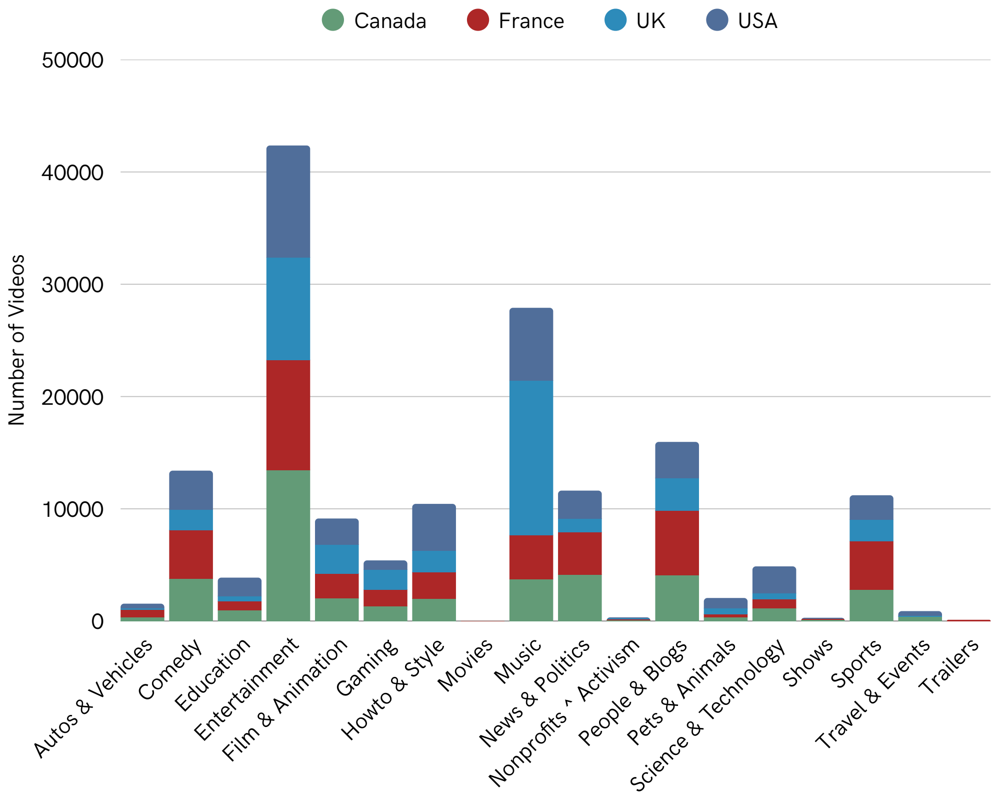
Key observations

- Average views are highest on Fridays, followed by Thursdays, while Saturdays receive the lowest view counts.

Business Insights

- End-of-week uploads (Thursday–Friday) maximize reach and visibility, whereas weekend releases - especially Saturday are less effective for driving views.

Entertainment and Music dominate the video count across all countries



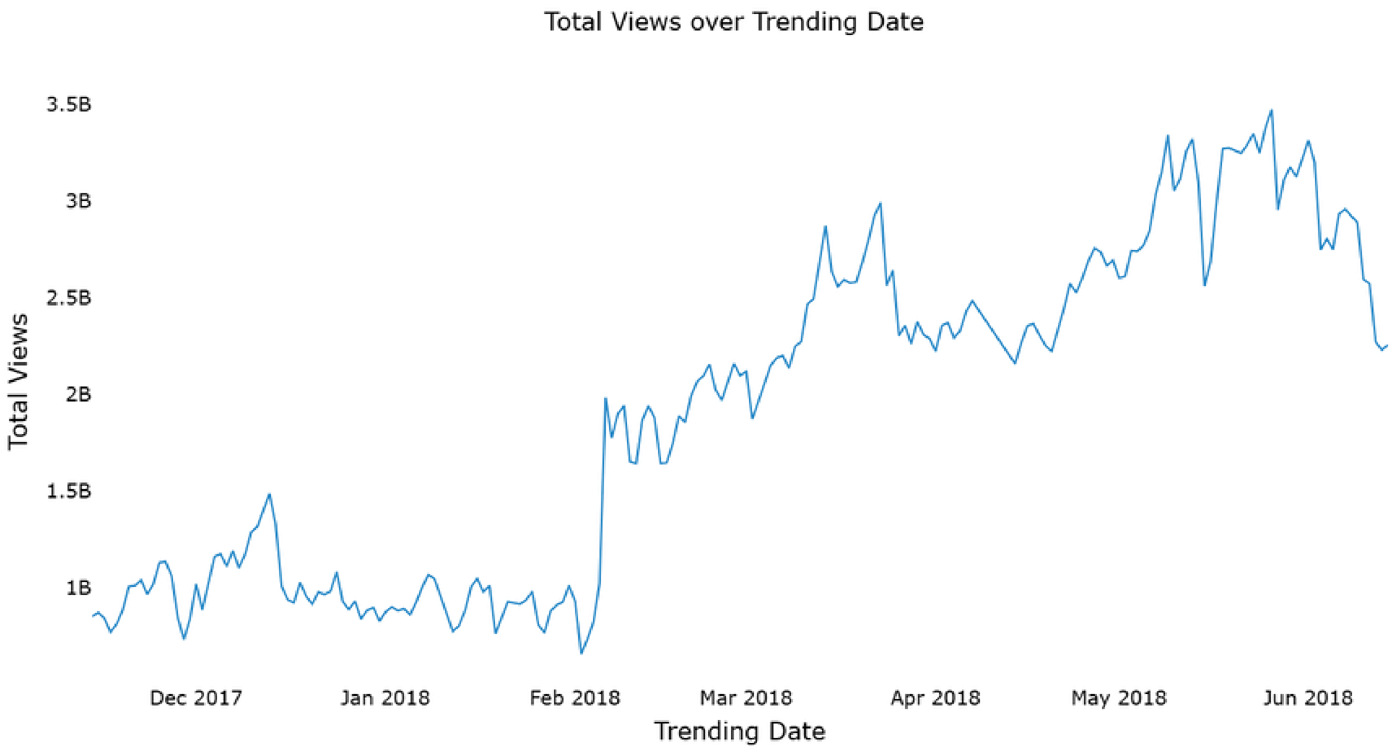
Key observations

- Entertainment has the highest video count in every country, led by the USA and Canada.
- Music is the second most dominant category, especially strong in the UK and USA.

Business Insights

- Global audience interest is heavily concentrated in Entertainment and Music, regardless of region.
- Country-specific strategies matter less for top categories but become important for niche ones like Education, Nonprofits, and Shows.

Total views on trending videos show a sustained upward trend with periodic spikes



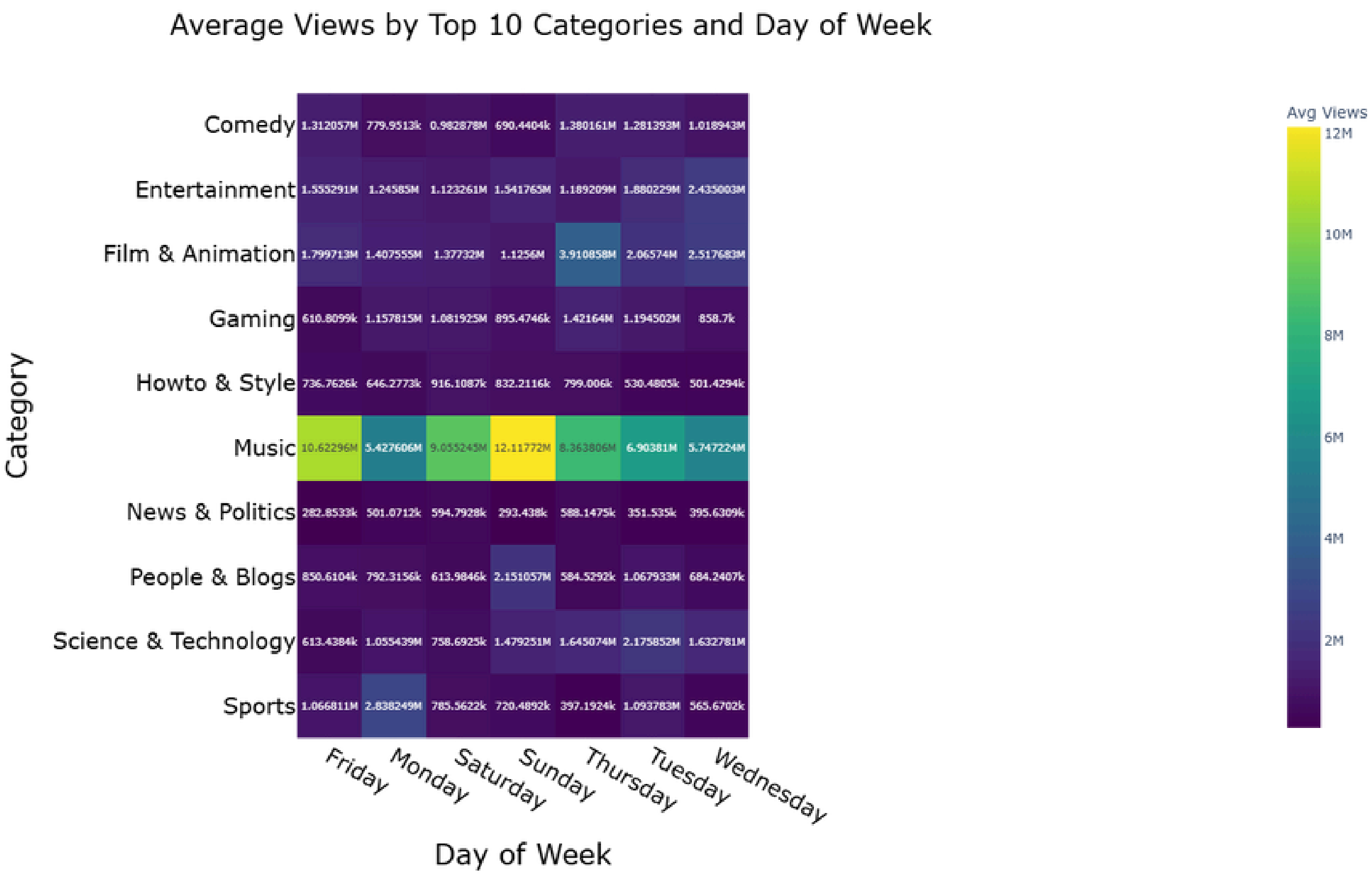
Key observations

- Total views increase steadily over time, with a sharp jump around early Feb 2018.
- Multiple spikes suggest viral events or high-impact content days rather than smooth growth.

Business Insights

- YouTube trending exposure is amplifying over time, increasing overall visibility potential.
- Timing content around high-engagement periods can significantly boost total views.

Music dominates average views across all days, with peak engagement on weekends



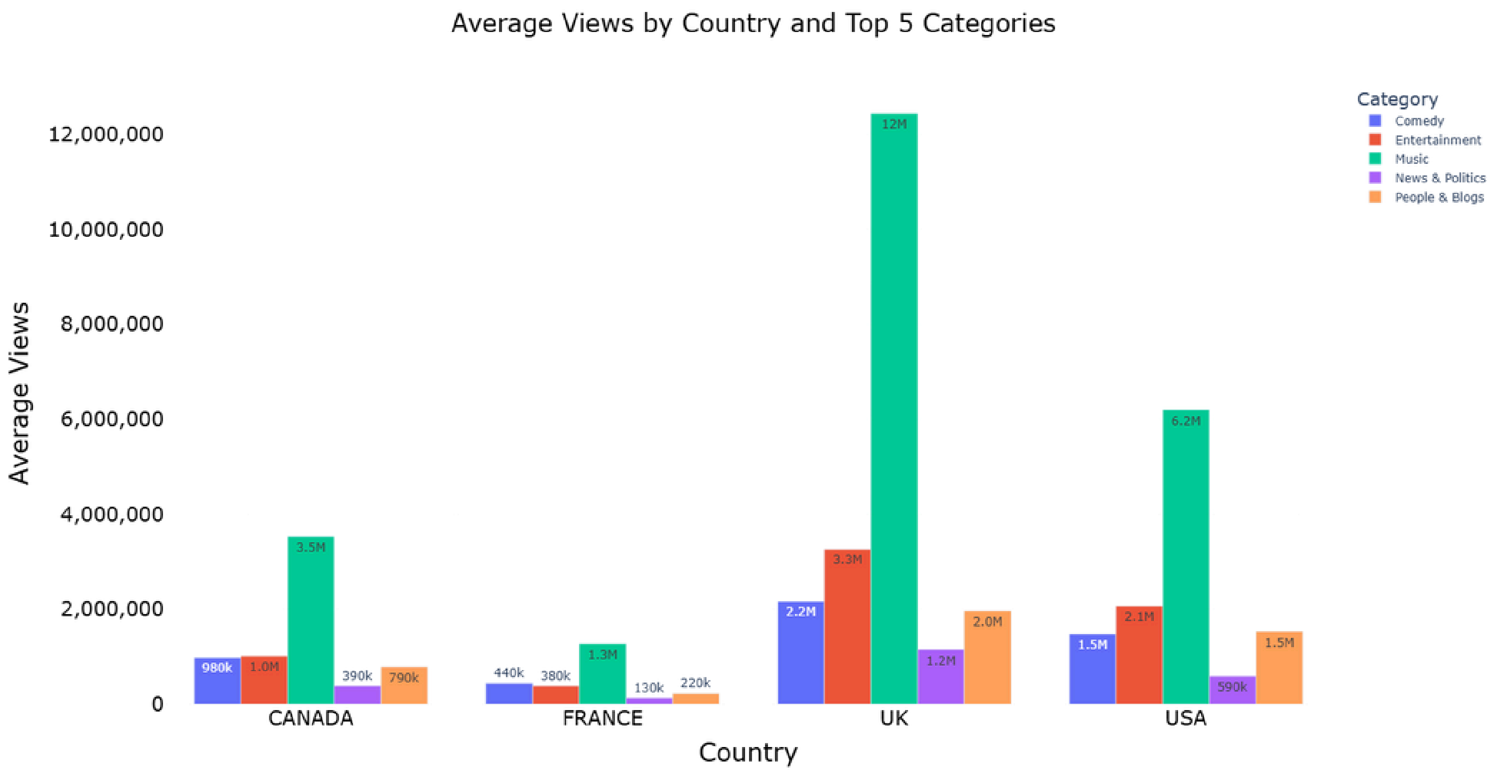
Key observations

- Music consistently outperforms all categories, especially on Saturday and Sunday.
- Most other categories show relatively stable but lower engagement across weekdays.

Business Insights

- Publishing music content on weekends maximizes average view potential.
- For non-music categories, day of upload matters less than content type.

UK videos achieve the highest average views, driven primarily by Music and Entertainment content



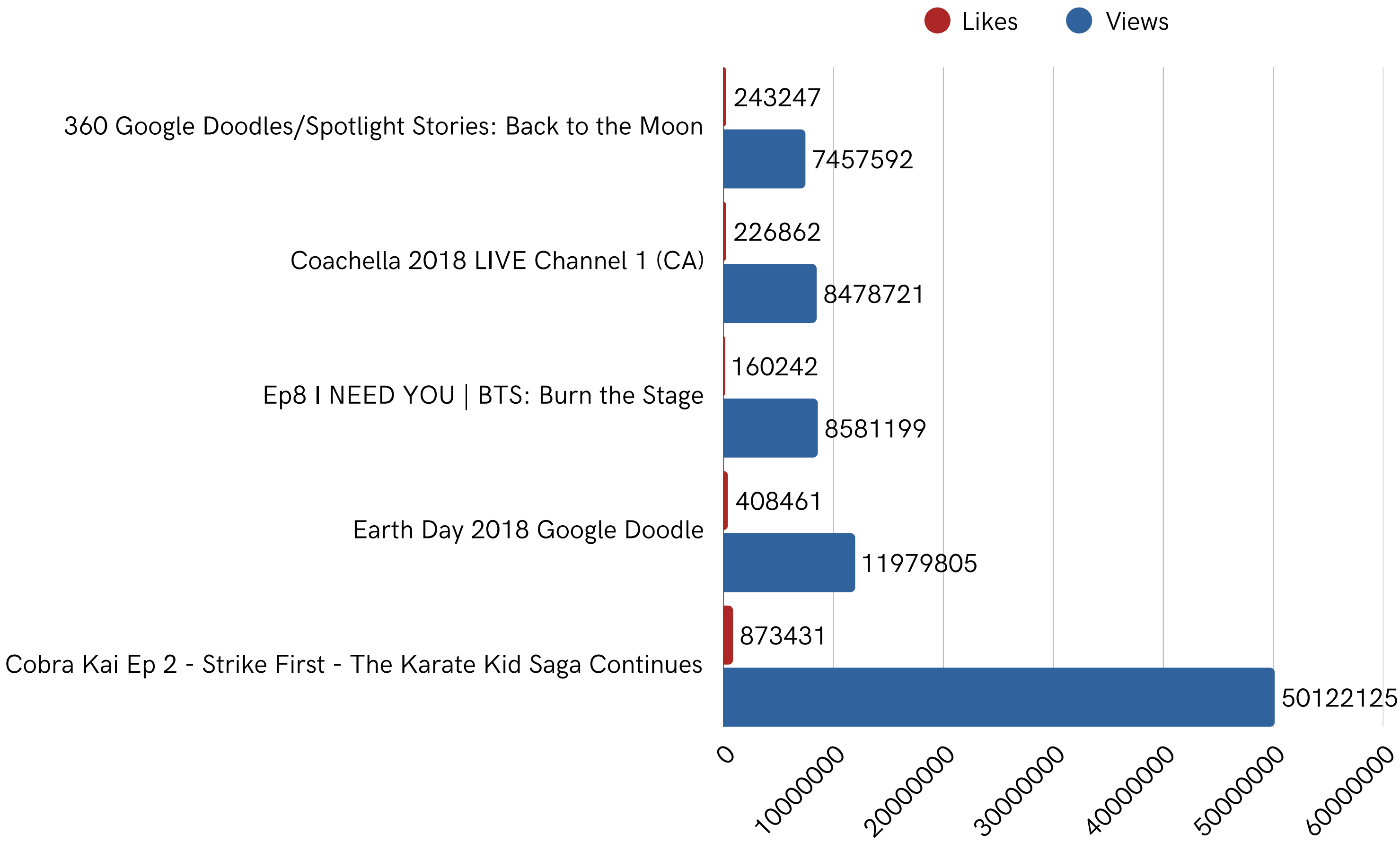
Key observations

- The UK significantly outperforms other countries in average views, especially in Music (~12M).
- Music leads in all countries, but the gap is widest in the UK, followed by the USA.

Business Insights

- The UK market shows the strongest engagement potential, particularly for music and entertainment creators.
- Content strategies should be region-specific, as the same categories perform very differently across countries.

High-Engagement Videos Can Be Removed Despite Massive Viewership



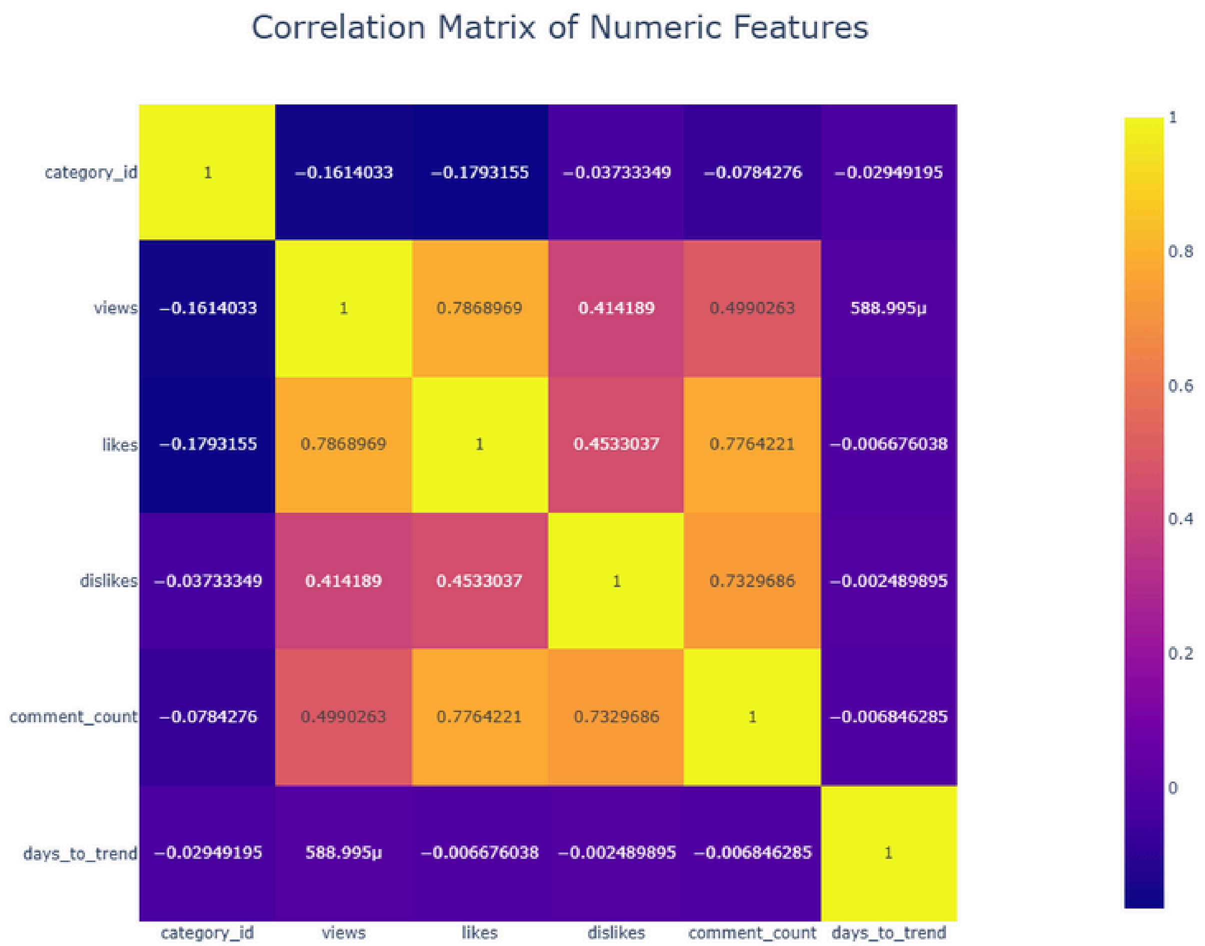
Key observations

- Several removed videos accumulated millions of views and substantial likes before removal, with some exceeding 50M views, indicating strong audience demand.

Business Insights

- High engagement does not guarantee content longevity.
- Policy enforcement, licensing or platform decisions can override popularity, highlighting the importance of compliance alongside reach.

User Engagement Metrics Are Strongly Interconnected, While Trending Speed Is Largely Independent



Key observations

- Views correlate strongly with likes (≈0.79) and moderately with comments and dislikes, showing that higher visibility drives overall engagement.
- Days to trend shows almost no correlation with any engagement metric, indicating timing has little impact on engagement levels.

Business Insights

- High engagement metrics (likes, comments, dislikes) tend to grow together as a video gains visibility.
- Speed to trend is driven more by platform algorithms or external factors than by raw engagement alone.

BUSINESS / DEVELOPER TAKEAWAYS

- **Established Channels Dominate:** Visibility is heavily concentrated among big channels. Smaller creators must use niche content or viral strategies to compete.
- **Timing & Engagement Windows:**
  - Uploading videos on Thursday–Friday maximizes reach and engagement.
  - Late-night/early-morning uploads can also improve visibility due to lower competition, while weekends generally underperform (except for Music).
- **Category Selection Drives Reach:** High-volume categories like Music face intense competition, while niche categories (Education, Science & Technology) offer opportunities for differentiation and audience growth.
- **Consistency Over Virality:**
  - Relying on rare viral hits is risky.
  - Consistent publishing and optimization improve long-term performance.

BUSINESS / DEVELOPER TAKEAWAYS

- **Early Performance is Critical:** Optimizing the first 24–48 hours (titles, thumbnails, timing, promotion) is far more impactful for trending success than long-term discovery.

- **Engagement Correlations:**
  - Views strongly correlate with likes and moderately with comments/dislikes.
  - High engagement metrics tend to grow together, but trending speed is driven more by platform algorithms or external factors than raw engagement.

- **Region-Specific Strategies Matter:**
  - Global participation is uniform, but country-specific optimization is key for niche categories.
  - The UK shows the strongest engagement potential, especially for Music and Entertainment.

- **Policy & Platform Factors Influence Longevity:**
  - High engagement doesn’t guarantee long-term visibility.
  - Platform policies, licensing and enforcement decisions can override popularity.

CHALLENGES & OPPORTUNITIES

Limitations

- The dataset is time bounded and may not reflect current trends or changes.
- Trending videos represent a small subset of all content, creating bias toward highly popular videos.
- Geographic coverage is limited to selected countries and may not capture global viewing behavior.
- Temporal relevance is constrained, as long-term performance and post-trending behavior are not captured.

Future Work / Opportunities:

- Perform sentiment analysis on video comments to understand audience perception and reactions.
- Conduct time-series analysis on views and likes to study how engagement evolves before and after trending.
- Compare trending videos with non-trending videos to identify factors driving trend eligibility.
- Analyze the impact of video duration, title keywords and thumbnails on engagement and reach.

CONCLUSION

This analysis of YouTube trending videos provided a comprehensive understanding of content trends, audience behaviour & performance characteristics. Key patterns and opportunities were identified, demonstrating how data driven insights can inform strategic decisions for content creators, marketers, and platforms. Future analysis could include sentiment analysis of viewer comments, time series tracking of engagement patterns, and comparisons with non trending or cross platform video data to uncover deeper insights.

This analysis was conducted using Python, Pandas, Matplotlib, Seaborn and Plotly.

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